Subject: Mathematics | **PERIODIC TEST – 03 (SURFACE AREAS & VOLUMES)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 13: Surface Areas and Volumes** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Section A has 4 MCQs and 1 Assertion-Reason question (1 mark each).
2. Section B has 2 VSAQ questions (2 marks each). Section C has 2 SA questions (3 marks each).
3. Section D has 1 Long Answer HOTS question (5 marks). Section E has 1 Applied Case Study (4 marks).
4. Take $\pi = \frac{22}{7}$ unless specified otherwise. Show clear steps in calculations.

SECTION A: OBJECTIVE & MCQs (1 Mark Each) [5 Marks]

1. A circus tent is cylindrical to a height of 4 m and conical above it. If its diameter is 105 m and its slant height is 40 m, the total area of the canvas required in m^2 is: **[1]**

(a) (a) 1760 (b) 2640 (c) 3960 (d) 7920

2. A sphere of radius 6 cm is dropped into a cylindrical vessel partly filled with water of radius 8 cm. When submerged completely, the water rises by: **[1]**

(a) (a) 4.5 cm (b) 3.5 cm (c) 4.0 cm (d) 2.0 cm

3. If a cone is cut into two parts by a horizontal plane passing through the mid-point of its axis, the ratio of the volumes of the upper part and the entire cone is: **[1]**

(a) (a) 1 : 2 (b) 1 : 4 (c) 1 : 6 (d) 1 : 8

4. The height of a cone is 30 cm. A small cone is cut off at the top by a plane parallel to the base of volume $\frac{1}{27}$ of the cone. The height above the base at which the section is made is: **[1]**

(a) (a) 10 cm (b) 15 cm (c) 20 cm (d) 25 cm

5. **Assertion (A):** Total surface area of a playing top formed by a cone mounted on a hemisphere is the sum of the curved surface area of the hemisphere and the curved surface area of the cone.
Reason (R): The top is obtained by fixing the flat circular surfaces of the hemisphere and cone together, making them internal. **[1]**

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

SECTION B: VERY SHORT ANSWER QUESTIONS (2 Marks Each) [4 Marks]

6. Two cubes have their volumes in the ratio 1 : 27. What is the ratio of their total surface areas? **[2]**

7. Two right circular cylinders of equal volumes have their heights in the ratio 1 : 2. What is the ratio of their radii? **[2]**

SECTION C: SHORT ANSWER QUESTIONS (3 Marks Each) [6 Marks]

8. A cylindrical container of radius 6 cm and height 15 cm is filled with ice-cream. The whole ice-cream is distributed to 10 children in equal cones with hemispherical tops. If the height of the conical portion is four times the radius of its base, find the radius of the ice-cream cone. **[3]**

9. A circus tent is cylindrical up to a height of 3 m and conical above it. If the diameter of the base is 105 m and the slant height of the conical part is 53 m, find the total canvas used in making the tent. **[3]**

SECTION D: LONG ANSWER / HOTS QUESTION (5 Marks) [5 Marks]

10. A building is in the form of a cylinder surmounted by a hemispherical vaulted dome and contains $41\frac{19}{21}$ m^3 of air. If the internal diameter of the building is equal to its total height above the floor, find the height of the building. (Take $\pi = \frac{22}{7}$) **[5]**

SECTION E: APPLIED CASE STUDY (4 Marks) [4 Marks]

11. Case Study: High-Precision Wooden Desk Pen Stand

A desk pen stand is manufactured from a block of wood shaped as a rectangular cuboid with four identical conical depressions drilled into the top surface to hold pens. Dimensions of the cuboid: Length = 15 cm, Breadth = 10 cm, Height = 3.5 cm. Dimensions of each conical depression: Radius $r = 0.5$ cm, Depth $h = 1.4$ cm.

(i) Find the volume of the original cuboidal wooden block before depressions were drilled.

(ii) Find the volume of each conical depression.

(iii) Find the total volume of wood remaining in the finished pen stand.

(iv) Find the total outer surface area of the cuboid.

[4]

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— END OF QUESTION PAPER —

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